Selection of Filler Composite Fabrication Techniques and Testing Methods for Characterization to Enhance Compression Properties of PTFE

Abstract

Pure polytetrafluoroethylene (PTFE) possesses excellent chemical inertness and a low coefficient of friction, yet its inherent cold flow (creep) tendency severely limits its use in mechanically loaded applications. Filler modification has been proven to be an effective strategy for enhancing the compression properties of PTFE; however, the selection of filler systems and the matching of fabrication-characterization methods determine the ultimate performance ceiling. This paper systematically reviews the selection framework for filler composite techniques to enhance PTFE compression performance, covering filler type and concentration optimization, composite process routes (cold pressing and sintering, liquid-phase mixing, compression molding), and compression characterization methodology. Based on literature data, filler concentrations in the range of 20–30 vol% can simultaneously achieve compression strength enhancement and processing feasibility, with compression strength improvements reaching 30–70% and hardness increases of 38–55%. A "filler-process-characterization" synergistic design principle is proposed to provide methodological guidance for the development of PTFE-based sealing and load-bearing components.

Keywords: Polytetrafluoroethylene; Filler modification; Compression properties; Cold pressing and sintering; Compression strength

1. Introduction

Polytetrafluoroethylene (PTFE) occupies an irreplaceable position in sealing, bearing, and sliding components due to its exceptional chemical inertness, wide temperature stability, and extremely low coefficient of friction. However, the mechanical deficiencies of pure PTFE are equally significant: its cold flow (creep) tendency causes irreversible deformation under sustained loads, and its compression strength is limited, severely constraining its application in high-pressure sealing and load-bearing conditions.

Since the 1960s, when filler modification was found to significantly improve the creep and wear properties of PTFE, filled PTFE has become the industrial standard for mechanical seals and bearing materials. However, filler modification is not a simple "add and improve" proposition—the matching relationships among filler type, concentration, composite process, and characterization methods determine the ultimate compression performance.

This paper systematically organizes the methodological framework for enhancing PTFE compression performance from the three dimensions of "filler selection—process matching—characterization verification," aiming to provide actionable decision-making guidance for materials engineers.

2. Filler System Selection Strategy

2.1 Classification and Functional Positioning of Fillers

PTFE fillers can be classified into three categories: **metallic fillers** (bronze powder, antimony powder, lead powder, molybdenum powder, nickel powder, etc.), **inorganic fillers** (graphite, molybdenum disulfide, glass fiber, carbon fiber, silicon dioxide, boron nitride, etc.), and **organic fillers** (polyarylsulfone, polyimide, polyphenylene sulfide, etc.).

From the perspective of enhancing compression performance, filler functions can be divided into two categories: **load-bearing fillers** share the load through their own high modulus, reducing the effective stress on the PTFE matrix; **lubricating fillers** improve tribological properties, indirectly suppressing compression softening caused by frictional heat.

Glass fiber is the most commonly used filler for improving PTFE compression performance and can effectively reduce PTFE creep at both low and high temperatures. Its chemical inertness makes it perform well in oxidizing environments, although attention must be paid to its adverse effect on dielectric breakdown strength (arising from increased porosity).

Carbon fiber possesses both high strength and high modulus, providing electrical conductivity to the composite while enhancing compression strength. Carbon fiber-filled modified PTFE compositions have been demonstrated to improve tensile elastic modulus and compression creep characteristics.

Inorganic nanofillers (such as SiO_2, ZrO_2, BN) can produce significant reinforcement effects at low loading levels. Research has demonstrated that introducing zirconium dioxide, silicon dioxide, and boron nitride nanofillers into carbon fiber-reinforced PTFE matrices results in **compression strength improvements of 30–70% and hardness increases of 38–55%**.

2.2 Optimal Window for Filler Concentration

There exists a clear effective window for filler concentration. Below 5 vol%, the effect of fillers on composite properties is insignificant; above 40 vol%, most physical properties drop sharply.

Table 1: Effect of Filler Concentration on PTFE Compression Properties

Filler Concentration	Effect	Applicability
< 5 vol%	Insignificant property improvement	Not recommended
20–30 vol%	Optimal balance of compression strength and processability	Recommended window
> 40 vol%	Sharp decline in physical properties	Avoid

The optimal filler content is typically **20–30 vol%**, depending on specific application requirements and filler combinations. Within this concentration range, compression strength increases, hardness increases, and creep resistance improves, while the reduction in tensile strength and elongation at break remains within acceptable limits.

2.3 Composite Filler Strategy

Single fillers often cannot simultaneously satisfy multiple performance requirements. Composite fillers achieve multi-objective performance optimization through "synergistic effects." Research has demonstrated that **vermiculite/kaolinite composite fillers** increase PTFE yield strength by 55%, compression strength by 60%, and reduce wear rate by a factor of 850, attributed to supramolecular reinforcement effects and the formation of hard tribofilms.

However, composite fillers do not always produce positive effects. Research on three-component fillers has shown **antagonistic effects** on PTFE wear performance, with results inferior to two-component fillers. This suggests that composite filler design must consider interactions between components rather than simple superposition.

3. Selection of Composite Process Routes

3.1 Cold Pressing and Sintering

Cold pressing and sintering is the most classic forming process for PTFE composites, with the flow sequence: **powder mixing → cold pressing into preform → high-temperature sintering → cooling**.

Cold pressing pressure is a key parameter affecting compression properties. Compression molding typically applies pressures ranging from **20 to 70 MPa** to compact the powder into a cohesive "green" part. A specific example from the patent literature describes a compression-molded sheet formed by fully compressing the powder at a pressure of approximately **30–60 MPa** (300–600 bar).

Sintering temperature is determined by the melting behavior of PTFE. The sintering process involves exposure to temperatures close to **400°C** for several hours. In practice, sintering is typically conducted at approximately **370–380°C** for several hours to ensure complete

particle fusion and elimination of porosity. A specific example describes a fully compressed preform sintered at about **370°C for approximately twenty-four hours**. Another example describes sintering at **370°C for 5 hours** with heating and cooling rates of 50°C/hour.

Research on sintering factors for PTFE-based materials has established that the optimal sintering parameters include a cooling rate of 25°C/h, sintering temperature of 360°C, and holding time of 6 hours. The cooling rate after sintering affects crystallinity and flexibility: rapid cooling produces lower crystallinity, favoring flexibility and bending resistance; slow cooling produces higher crystallinity.

3.2 Liquid-Phase Mixing

The extreme chemical inertness and non-stick nature of PTFE make uniform mixing with fillers difficult. Liquid-phase mixing improves mixing uniformity by dispersing PTFE powder and filler separately in solvents to form emulsions, followed by mixing, filtration, and drying. The advantage of liquid-phase mixing lies in the more uniform distribution of filler on the surface of PTFE particles, reducing local stress concentration caused by filler agglomeration.

3.3 Compression Molding and Multi-Layer Structure Design

For applications requiring both a low-friction surface and high load-bearing capacity, **multi-layer compression molding** provides an effective solution. Patent EP0706446B1 describes a dual-layer structure: the bottom layer is filled PTFE (containing 20% carbon filler), providing creep resistance; the top layer is virgin or modified PTFE, providing a low-friction surface.

The key to this process lies in **stepwise compression**: first, the first layer is partially preformed using a very low pressure of about **2.7 kPa** (0.027 bar, 0.4 psi). Next, a layer of virgin modified powdered PTFE resin is added to the compression mold cavity, and this layer and the first layer are fully compressed at a pressure of about **30–60 MPa**.

The resulting article provides a layer of low-friction modified PTFE while also providing improved resistance to creep through the layer of filled PTFE. The filled PTFE comprises carbon filler material at about **10 to 30 percent by weight** of the filled layer.

3.4 Comparison of PTFE Molding Techniques

The most commonly used PTFE molding techniques include compression molding, ram extrusion, and paste extrusion/injection.

Compression molding is the traditional and most widely used PTFE molding technique. It produces dense, high-strength components with excellent mechanical properties and is compatible with filled PTFE grades including glass, carbon, and bronze. Compression molding exhibits minimal shrinkage compared to extrusion techniques and achieves high dimensional accuracy.

Ram extrusion is used to produce continuous shapes like rods, tubes, and sheets. In this method, PTFE powder is mixed with a temporary lubricant to form a paste that is forced through a die. The resulting part generally has uniform density and consistent mechanical properties along its length. Ram extrusion is compatible with filled PTFE, including glass, carbon, and bronze reinforcements.

Paste extrusion or injection is a specialized process for producing complex, thin-walled, or delicate components. It enables complex shapes that are impossible with compression molding but may result in slightly lower mechanical properties compared to compression-molded parts.

4. Characterization Methods for Compression Properties

4.1 Compression Strength Testing

Compression strength is the core metric for evaluating the effectiveness of filler modification. The standard test methods are **ASTM D695** (Standard Test Method for Compressive Properties of Rigid Plastics) or equivalent standards such as GB/T 1041. Typical test parameters include: compression modulus (target value ≥ 500 MPa at room temperature), compression strength, and compression set.

Key detection items include: yield strength (0.2% offset yield strength, Rp0.2, per ASTM D695), ultimate compression strength (maximum load, σmax, in MPa), and elastic modulus (slope of initial linear region, E, ≥ 1000 MPa).

Research provides a quantitative benchmark for compression performance improvement: nanofiller-reinforced PTFE exhibits **compression strength improvements of 30–70%**.

4.2 Creep and Compression Set Testing

The cold flow behavior of PTFE is central to compression performance evaluation. **Compression creep testing** measures strain evolution over time under constant load and temperature, directly reflecting the filler's effectiveness in suppressing cold flow. **Compression set testing** (ASTM D395) evaluates recovery capability after unloading, which is particularly important for sealing applications.

Relevant detection items include: creep rate (deformation over time under constant load, CR, ≤ 0.01%/h) and creep rupture time (failure period under long-term load, tf, per ASTM D2990).

4.3 Hardness Testing

Hardness is an indirect indicator of compression performance. **Shore D hardness** is the most commonly used method, with typical target ranges of **70–95 D** for filled PTFE. Hardness improvement magnitude (38–55%) reflects the enhancement of local compressive resistance by fillers.

4.4 Microstructural Characterization

The mechanisms of compression performance enhancement need to be understood through microstructural analysis. **Scanning Electron Microscopy (SEM)** is used to observe filler distribution, interfacial bonding, and fracture surface morphology. Samples are cryo-fractured after soaking in liquid nitrogen for one minute, then observed with SEM at different magnifications.

Differential Scanning Calorimetry (DSC) determines crystallinity, which affects compression modulus and creep behavior. The heat of fusion is measured using DSC, with the sample first heated to 250°C, maintained for 1 minute, then heated to 380°C for complete melting, followed by cooling at 10°C/minute, and the heat of fusion calculated based on the thermal peak at crystallization.

Fourier Transform Infrared Spectroscopy (FTIR) can detect chemical interactions between fillers and PTFE, although the chemical inertness of PTFE typically results in weak interfacial chemical interactions.

4.5 Compression Deformation Analysis

For filled PTFE, the addition of fillers significantly affects the compaction behavior. The applied stress required to reach a given relative density is **higher** when fillers are added compared to virgin PTFE. The ratio of transmitted to applied stress decreases consistently with the addition of fillers, consistent with a higher internal friction coefficient and measured higher level of radial stress.

Research on the influence of fillers on compression deformation has shown that at compression by 30 MPa pressure, 10 mass% filled composites based on B_4C and Co exhibit deformation of 3.20% and 2.46% respectively, which is 6.5 and 8.5 times that of unfilled PTFE.

5. Synergistic Design Framework

Based on the above analysis, the "filler-process-characterization" synergistic design for enhancing PTFE compression performance should follow these principles:

Step 1: Filler Selection. Determine filler type based on compression performance requirements. For load-bearing priority, select glass fiber or carbon fiber; for multifunctional requirements, consider nanofiller composite systems; note the potential antagonistic effects of three-component fillers.

Step 2: Concentration Optimization. Control filler content at **20–30 vol%**; within this window, compression strength improvement and processing feasibility reach a balance.

Step 3: Process Matching. Use cold pressing pressure starting from 20–70 MPa for optimization; sintering temperature 370–380°C; sintering time 5–24 hours; liquid-phase mixing superior to dry mixing.

Step 4: Characterization Verification. Compression strength testing verifies core performance; creep and compression set testing evaluate long-term load-bearing capacity; hardness testing serves as a rapid screening indicator; SEM/DSC reveal structure-property relationships.

6. Conclusions

The methodological core for enhancing PTFE compression performance lies in the synergistic selection of filler systems, composite processes, and characterization methods. The filler concentration of 20–30 vol% represents the optimal balance between compression strength and processability. Glass fiber and carbon fiber are the mainstream choices for enhancing compression load-bearing capacity. The introduction of nanofillers can achieve 30–70% compression strength improvement at low loading levels. Cold pressing and sintering (pressure

20–70 MPa, sintering temperature 370–380°C) is the process route with the highest industrial feasibility, and liquid-phase mixing can further improve dispersion uniformity. Compression performance verification should center on compression strength testing, supplemented by creep and hardness testing, combined with microstructural characterization to understand the reinforcement mechanisms.

References

[1] Filled Polytetrafluoroethylene, Chapter 12.1 Introduction, ScienceDirect.

[2] Method for preparing high thermal conductivity polytetrafluoroethylene composite membrane material, CN109181178A.

[3] Modified PTFE molded article, EP1829904A1.

[4] Articles comprising filled PTFE, US5399307.

[5] Effect of Sintering Factors on Properties of Al-Rich PTFE/Al/TiH2 Active Materials, Polymers 13 (2021) 1705.

[6] PTFE Compression Strength Testing, Yjsyi Testing Center, 2025.

[7] Study on Mechanical Properties of PTFE Modified by Polybenzoate and Carbon Fiber, Lubrication and Sealing, 2024(11).

[8] Differences between PTFE molding techniques, Felis S.r.l., 2026.

[9] Composite Systems made of PTFE for sealing and engineered solutions, ElringKlinger Kunststoff.

[10] Modeling of the mechanical behavior of PTFE compounds during their cold pressing, HAL Theses.

[11] Investigation of the Tribological Properties of Polytetrafluoroethylene Modified with Metallic and Nonmetallic Nano Particles, Semantic Scholar, 2015.

[12] Preparation and Mechanical Properties of Nano-Al_2O_3/PTFE Composites, Wanfang Data.

[13] Anisotropic deformation of PTFE during sintering, HAL Science.

[14] The 3D compaction tool allows to applied complex loadings to PTFE samples, HAL Theses.

[15] Non-linear DMA results of samples produced with different sintering cycles, HAL Theses.

[16] Mechanics of filled PTFE composites, Vestnik DonSTU.